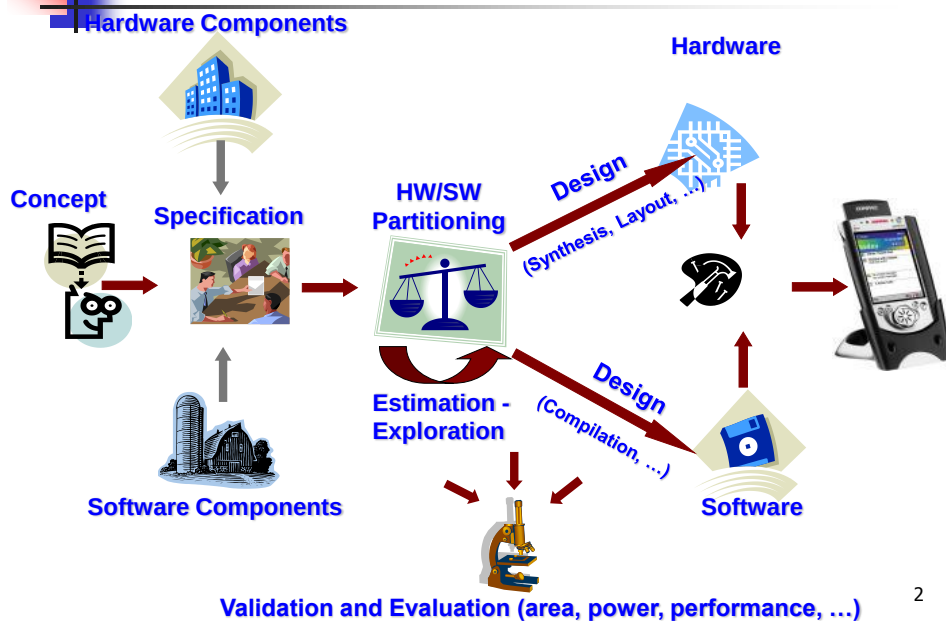


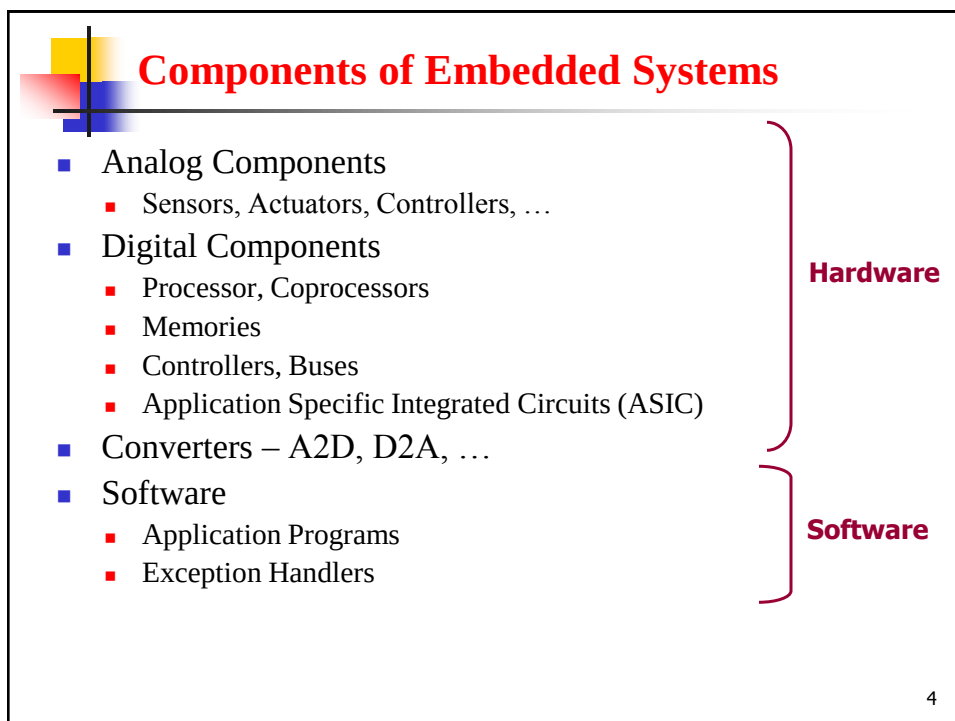
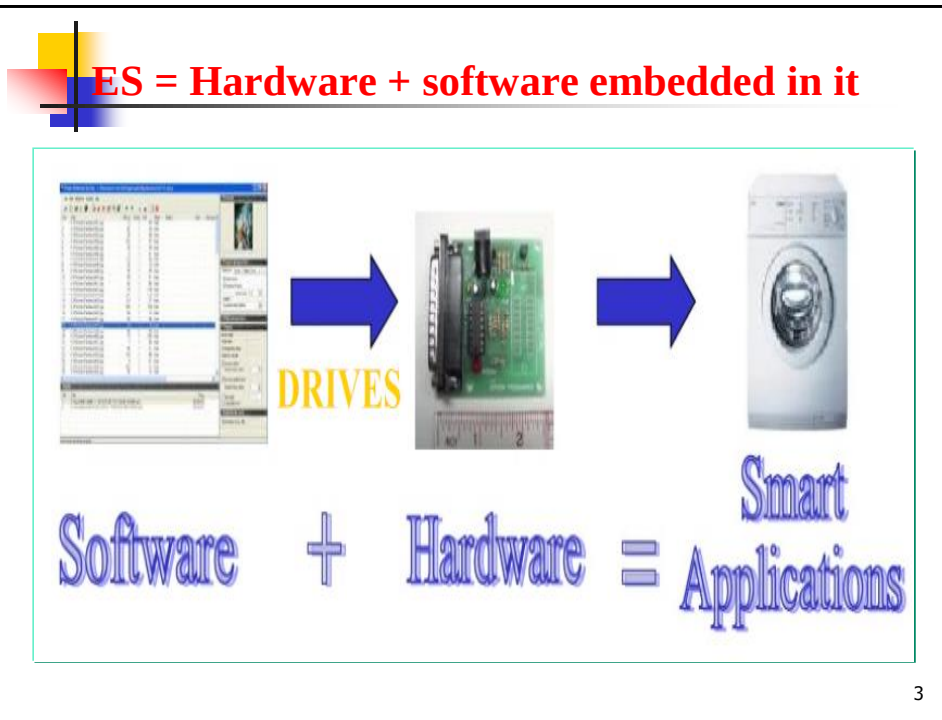
Chapter 2

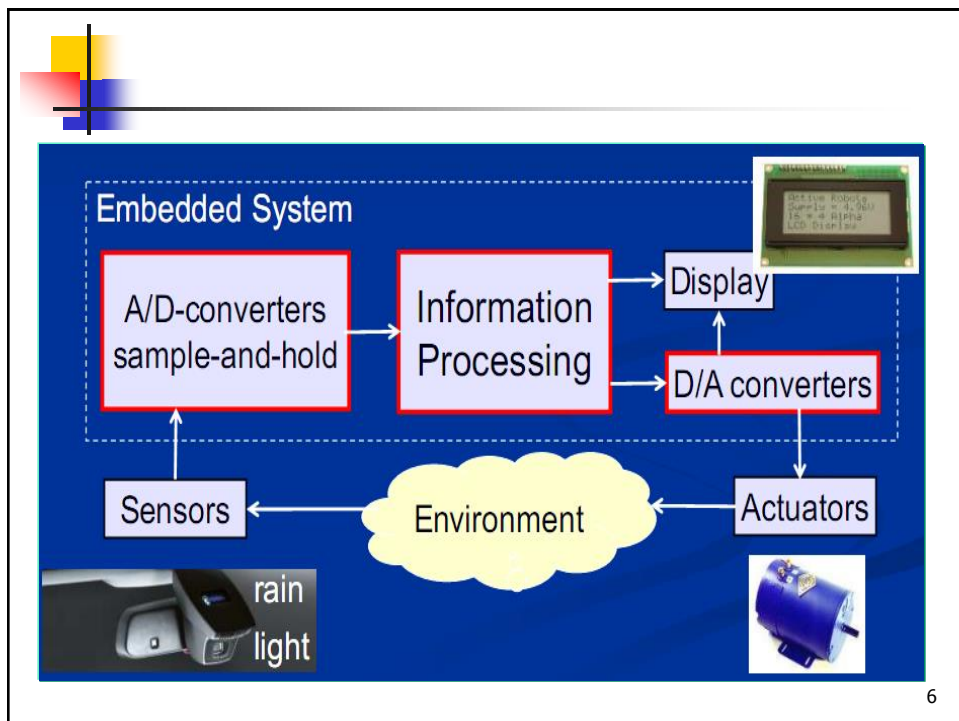
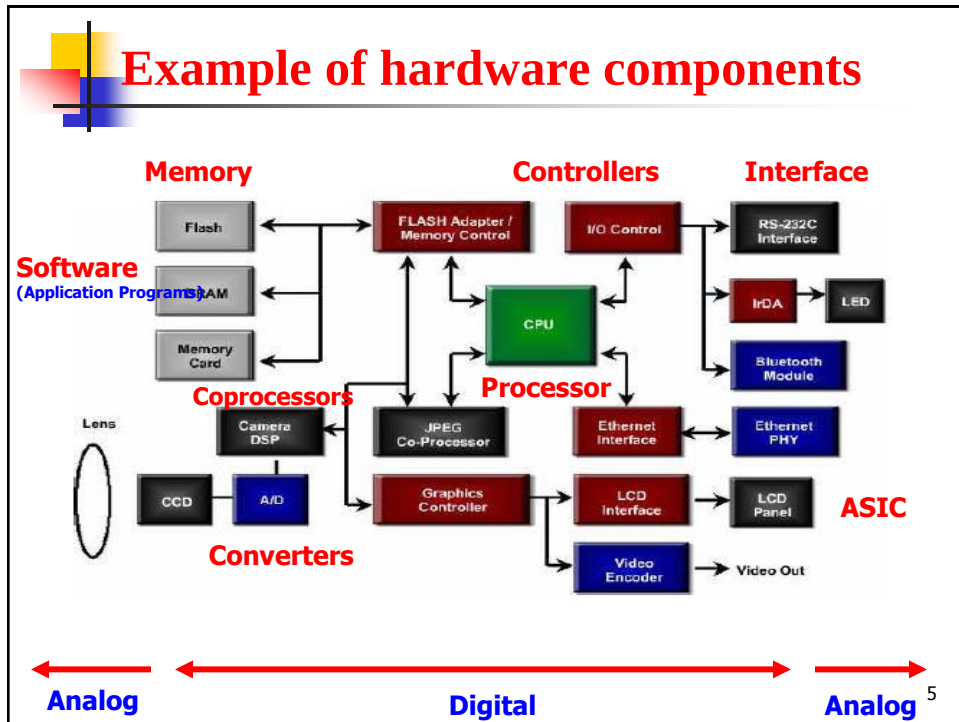
COMPONENTS OF EMBEDDED SYSTEMS

1

Course Outline







Sensors

- Processing of physical data starts with capturing this data.
- Sensors can be designed for virtually every physical and chemical quantity (hầu hết cho các đại lượng hóa lý)
 - including weight, velocity, acceleration, electrical current, voltage, temperatures etc.
 - chemical compounds.
- Many physical effects used for constructing sensors.
- Examples:
 - law of induction (generation of current in a magnetic field),
 - light-electric effects
- Huge amount of sensors designed in recent years.

7

Examples of Sensors



8



Signals

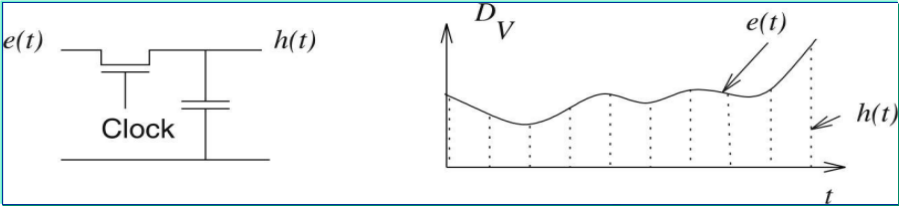
- Sensors generate signals
 - Definition: a signal s is a mapping from the time domain DT to a value domain DV :
 $S: DT \rightarrow DV$
 - DT : continuous or discrete time domain
 - DV : continuous or discrete value domain
- Digital computers require discrete sequences of physical values, i.e., DT should be discrete time domain.
- Digital computers require digital/binary form of physical values, i.e., Dv should be discrete value domain.
- How to do this for continuous physical signals?
 - Sample-and-hold circuits (discretization in time or sampling)
 - A/D-converters (discretization of values or quantization)

9



Discretization in Time: Sample-and-hold circuit

- Clocked transistor + capacitor;
- Capacitor stores sequence values

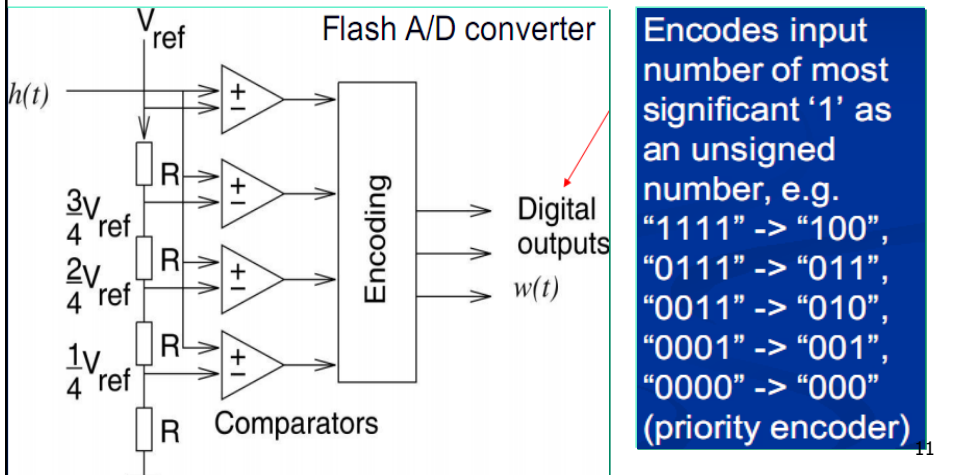


- $e(t)$ is a mapping $\mathbb{R} \rightarrow \mathbb{R}$
- $h(t)$ is a sequence of values or a mapping $\mathbb{Z} \rightarrow \mathbb{R}$

10

Discretization of Values:

- Discretization of Values: Analog-to-Digital (A/D)-converters (1)



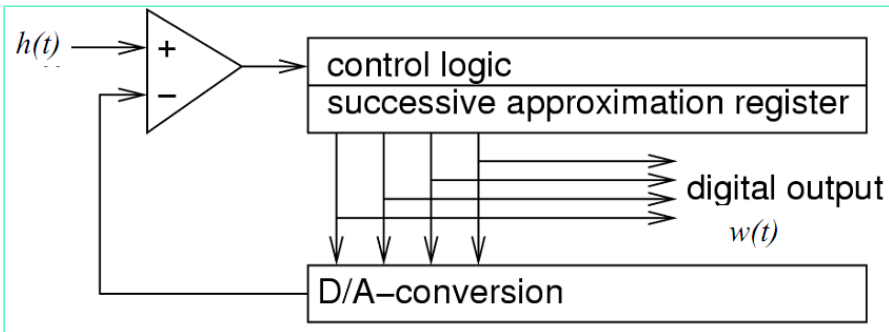
Resolution and Speed of Flash A/D converter

- Resolution (in bits): number of bits produced
- Resolution Q (in volts): difference between two input voltages causing the output to be incremented by 1

$$Q = \frac{V_{FSR}}{n}$$

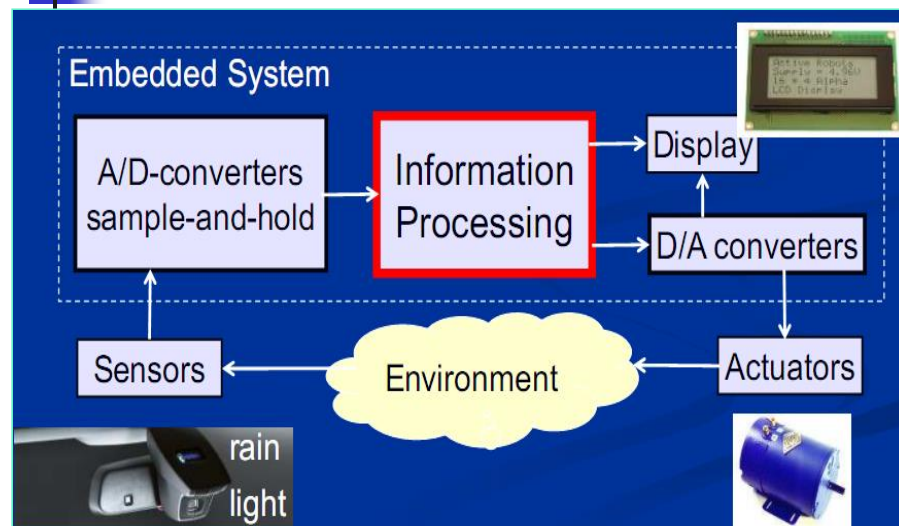
V_{FSR} : difference between largest and smallest voltage
 n : number of voltage intervals

■ Discretization of Values: A/D-converters (2)



Key idea: binary search:
 Set MSB = '1'
 if too large: reset MSB
 Set MSB - 1 = '1'
 if too large: reset MSB-1

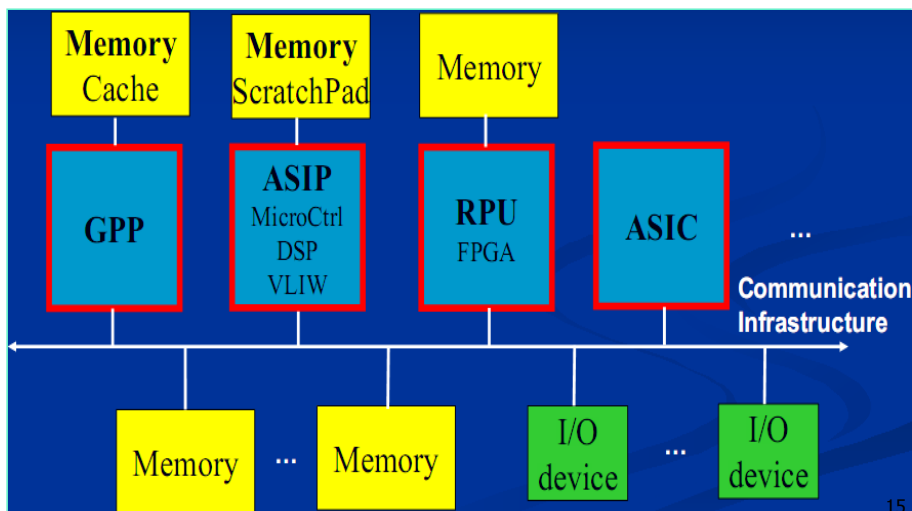
13



14

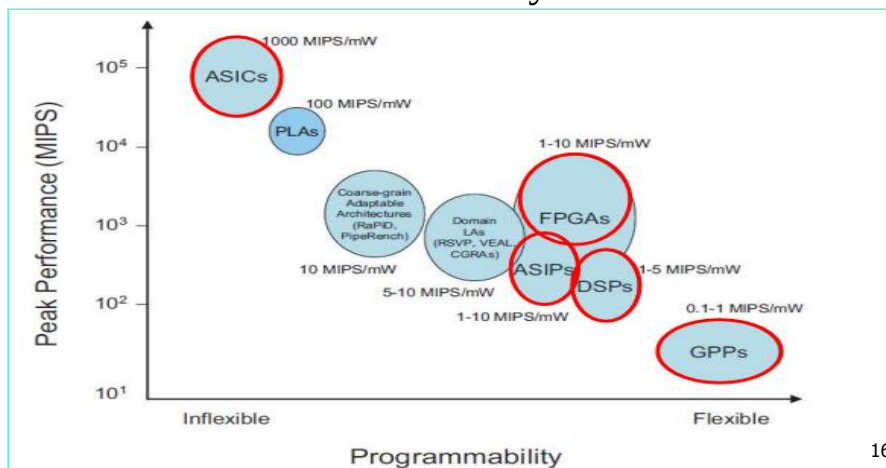
Information Processing System:

■ Computation Components



Why Implementation Alternatives?

■ Trade-off between Flexibility and Performance/Power Efficiency

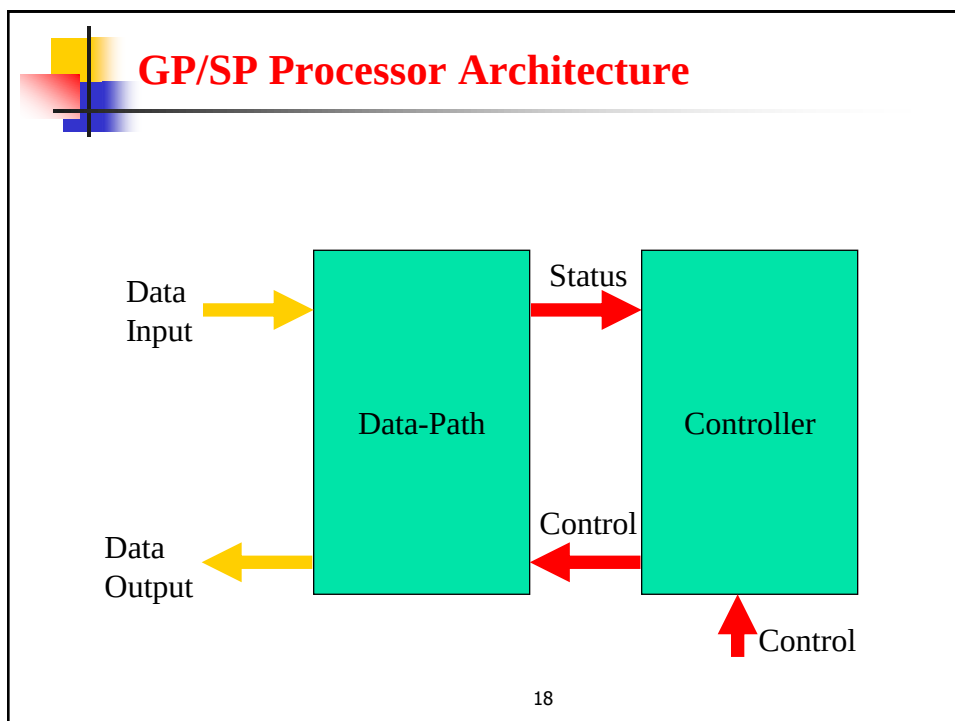




Processors

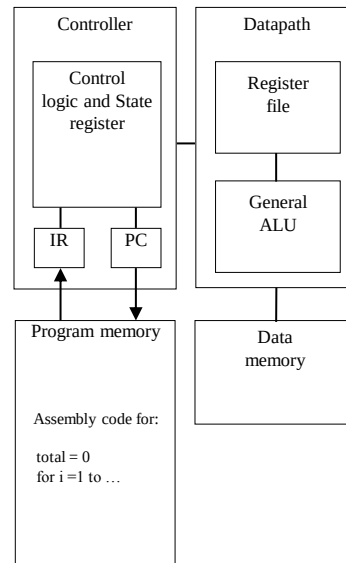
- What is a processor?
 - Artifact (công cụ) that computes (runs algorithms)
 - Controller and data-path
- General-purpose (GP) processors:
 - Variety of computation tasks
 - Functional flexibility and low cost at high volumes (maybe)
 - Slow and power hungry
- Single-purpose (SP) processors (or ASIC)
 - One particular computation task
 - Fast and power efficient
 - Functional inflexibility and high cost at low volumes (maybe)

17



General-purpose processors

- Programmable device used in a variety of applications
 - Also known as “microprocessor”
- Features
 - Program memory
 - General datapath with large register file and general ALU
- User benefits
 - Low time-to-market
 - High flexibility
- Examples
 - Pentium, Athlon, PowerPC



19

General Purpose Processors (GPPs)

- Can achieve high performance
 - Highly optimized circuits
 - Use of instruction-level parallelism
 - superscalar: dynamic scheduling of instructions
 - super-pipelining: instruction pipelining, branch prediction, speculation
 - Complex memory hierarchy – caches
- Not suitable for real-time applications
 - Execution times are highly unpredictable because of intensive resource sharing and dynamic decisions
- Properties
 - Good average performance for large application mix
 - High power consumption

20



ASIP

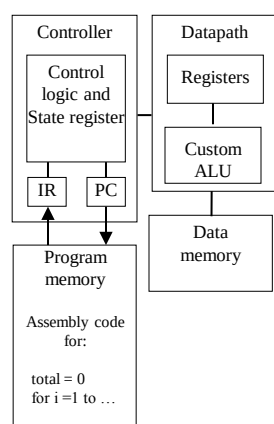
- Application Specific Instruction Set Processors (ASIPs)
- Micro Controllers (MicroCtrl)
 - Used in Control Dominated Systems (HT điều khiển chính)
 - Reactive systems with event driven behavior
 - Application examples: cars, consumer electronics (washing machines, dishwashers etc.)
- Digital Signal Processors (DSPs)
 - Used in Data Dominated Systems
 - Streaming-oriented systems with mostly periodic behavior
 - Application examples: signal processing
- Very Long Instruction Word Processors (VLIWs)
 - Used in Data Dominated Systems
 - Application examples: image processing

21



Application-specific IS processors (ASIPs)

- Programmable processor optimized for a particular class of applications having common characteristics
 - Compromise between general-purpose and ASIC (custom hardware)
- Features
 - Program memory
 - Optimized datapath
 - Special functional units
- Benefits
 - Some flexibility, good performance, size and power
- Examples
 - DSPs, Video Signal Processors, Network Processors,...

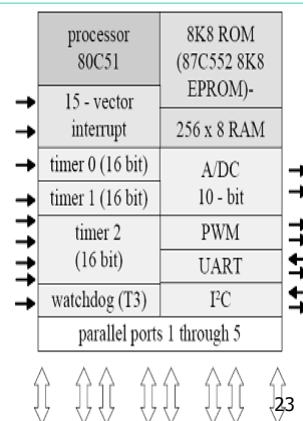


22



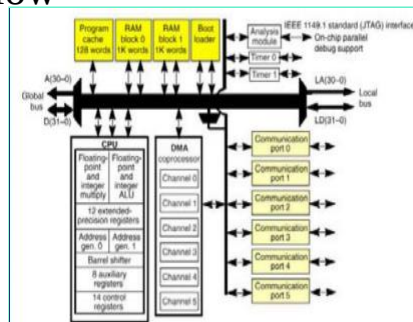
Micro Controllers

- Control-dominant applications (UD thiên về ĐK)
 - Supports process scheduling and synchronization
 - Preemption (interrupt), context switch
 - Short latency times
- Low power consumption
- Peripheral units often integrated
- Suited for real-time applications



Digital Signal Processors

- Optimized for data-flow applications
- Suited for simple control flow
- Parallel hardware units
- Specialized instruction set
- High data throughput
- Zero-overhead loops
- Specialized memory
- Suited for real-time applications



Reconfigurable Processing Units (RPU)

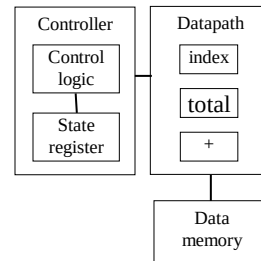
- Full custom circuit (HW) may be too expensive, software (SW) too slow.
- Combine the speed of HW with the flexibility of SW
 - HW with programmable functions and interconnect.
 - HW (Re-)Configurable at design-time or at run-time (dynamic reconfiguration)
- Field Programmable Gate Arrays (FPGAs)
 - Currently the most sophisticated and used RPUs
 - Applications
 - Fast and very cheap prototyping of (MP-)SoCs
 - Encryption,
 - Fast “object recognition” (medical and military)
 - Adapting mobile phones to different standards
- Very popular devices from
 - XILINX (Virtex II(Pro), Virtex 4, Virtex 5, Virtex 6, Virtex 7)₂₅
 - Altera, Actel and others

Application Specific Integrated Circuits (ASICs)

- Custom-designed circuits necessary
 - if ultimate speed or energy efficiency is the goal and large numbers can be sold. (nếu tốc độ, hiệu quả năng lượng, là mục tiêu cuối cùng và số lg có thể bán)
- Approach suffers from
 - long design times,
 - lack of flexibility (changing standards)
 - high costs, i.e., Millions of \$ mask costs

Application-Specific ICs (ASICs)

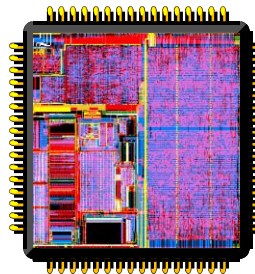
- Digital circuit designed to execute exactly one program
 - coprocessor, hardware accelerator
- Features
 - Contains only the components needed to execute a single program
 - No program memory
- Benefits
 - Fast
 - Low power
 - Small size



27

Application Specific Circuits (ASIC)

- Custom-designed circuits necessary if ultimate speed or energy efficiency is the goal and large numbers can be sold.
- Approach suffers from long design times and high costs.



28



GP vs. SP Processors

GP: ■ Programmable controller ■ Control logic is stored in memory ■ Fetch/decode overhead ■ Highly general data-path ■ Typical bit-width (8, 16, 32, 64) ■ Complete set of arithmetic/logic units ■ Large set of registers ■ High NRE/sale-volume	ASIC: ■ Hardwired controller ■ No need for program memory and cache ■ No fetch/decode overhead ■ Highly tuned data-path ■ Custom bit-width ■ Custom arithmetic/logic units ■ Custom set of registers ■ Low NRE/sale-volume
--	---

29



Storage

- What is a memory?
 - Artifact that stores bits
 - Storage fabric and access logic
- Write-ability
 - Manner and speed a memory can be written
- Storage-permanence
 - ability of memory to hold stored bits after they are written
- Many different types of memories
 - Flash, SRAM, DRAM, etc.
 - Common to compose memories

30



Write-ability

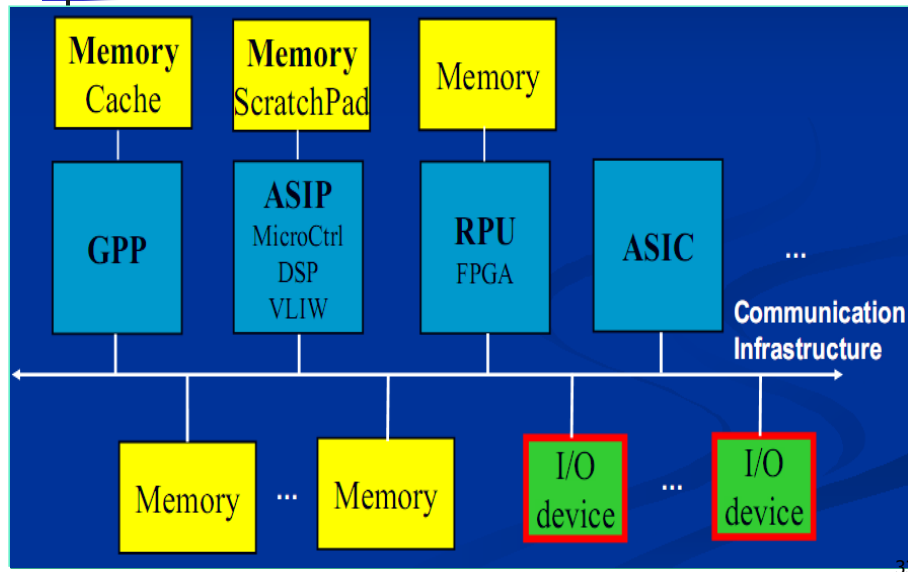
- Ranges of write ability
 - High end (cao cấp)
 - Processor writes to memory simply and quickly
 - E.g., RAM
 - Middle range
 - Processor writes to memory, but slower
 - E.g., FLASH, EEPROM
 - Lower range
 - Special equipment, “programmer”, must be used to write to memory
 - E.g., EPROM, OTP ROM
 - Low end
 - Bits stored only during fabrication
 - E.g., Mask-programmed ROM₃₁



Storage-permanence (lưu trữ lâu dài)

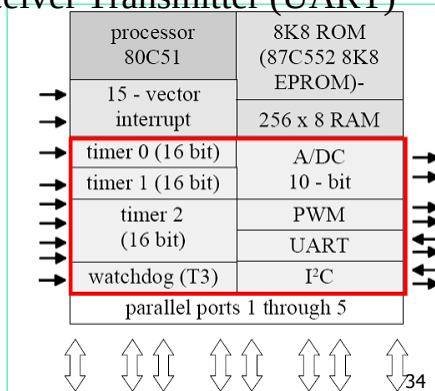
- Range of storage permanence
 - High end
 - Essentially never loses bits
 - E.g., mask-programmed ROM
 - Middle range
 - Holds bits days/months/years after memory's power source turned off
 - E.g., NVRAM
 - Lower range
 - Holds bits as long as power supplied to memory
 - E.g., SRAM
 - Low end
 - Begins to lose bits almost immediately after written
 - E.g., DRAM

Input/Output Devices

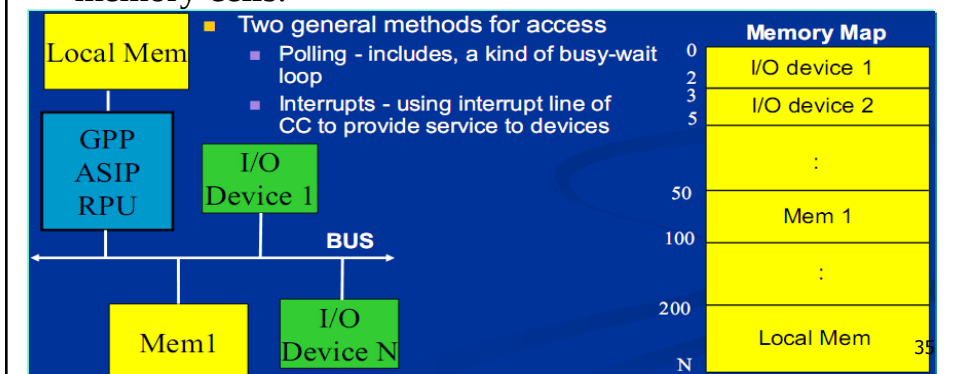


Examples of I/O Devices

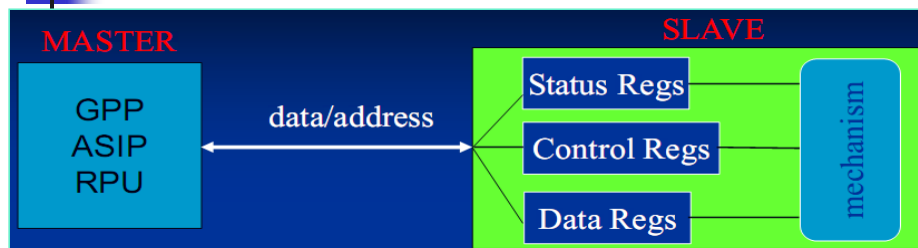
- Timers
- WatchDogs
- Pulse Width Modulators (PWM)
- Universal Asynchronous Receiver Transmitter (UART)
- Inter Integrated Circuit (I2C)
- ...



- How Computation Components “talk” to I/O Devices?
- I/O Devices are mapped in the address space of Computation Components (CC), i.e., CCs communicate with I/O devices simply by reading/writing from/to memory cells.



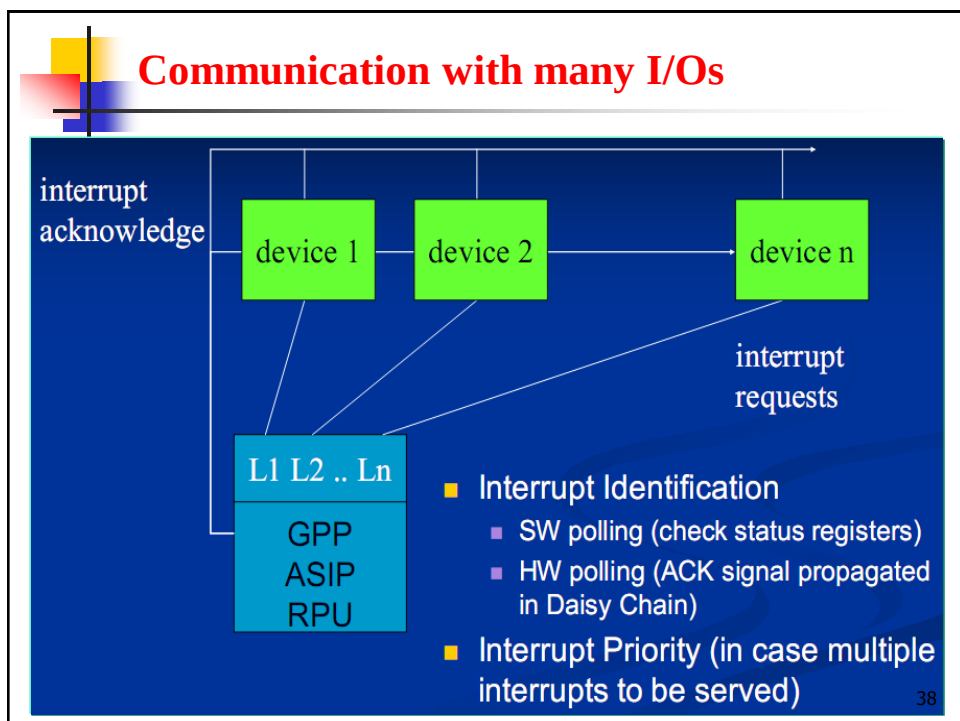
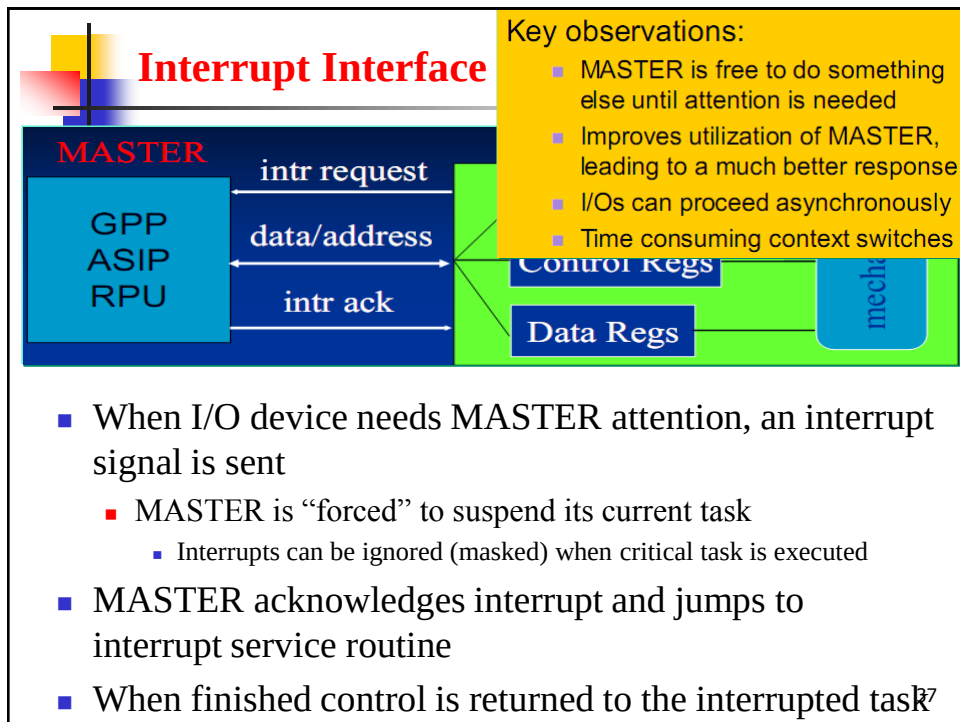
Polling: Busy-Wait Interface



- Continuous polling
 - Loop (until device ready)
 - Check device
 - End-loop
- Periodic polling

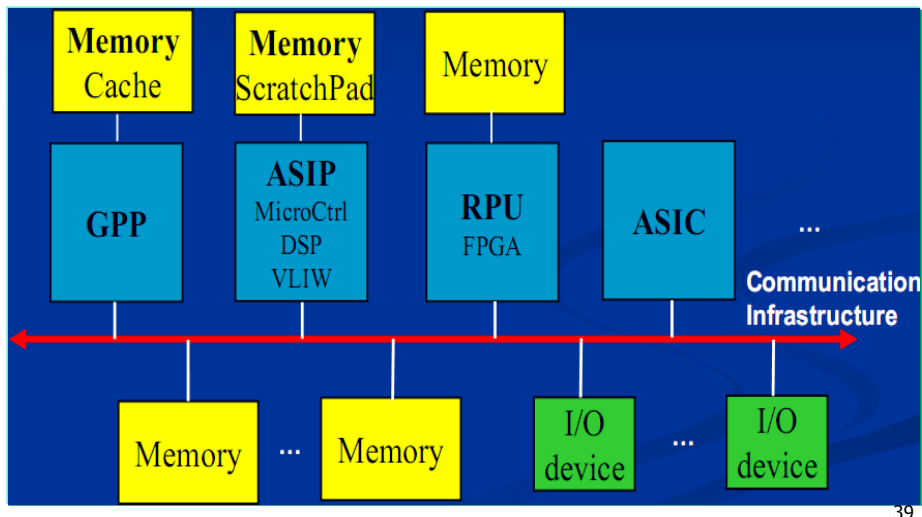
Issues:

- MASTER is tied up in communication with device until I/O operation is done
- No other work can be accomplished by MASTER
- Typically one MASTER in system, but many I/O devices
- Only really useful if devices are fast
 - No time consuming context switches



Information Processing System:

■ Communication Infrastructure



Types and Requirements

■ Types of Communication Infrastructure

- Single Bus
 - Single shared resource, thus may become the bottleneck in the system
- Multiple Busses/Crossbar Switches
 - Wire Switching communication infrastructure
- Networks (on Chip)
 - Packet Switching communication infrastructure

■ Requirements for Embedded Systems

- Performance
 - bandwidth and latency
 - guaranteed behavior (real-time)
- Efficiency
 - cost (material, installation, maintenance)
 - low power
- Robustness (mạnh mẽ)
 - fault tolerance
 - maintainability, diagnoseability, security, safety



Communication

- What is a bus?
 - An artifact that transfers bits
 - Wires, air, or fiber and interface logic
- Associated with a bus, we have:
 - Connectivity scheme
 - Serial Communication
 - Parallel Communication
 - Wireless Communication
 - Protocol
 - Ports
 - Timing Diagrams
 - Read and write cycles
 - Arbitration scheme, error detection/correction, DMA, etc.

41



Serial Communication

- A single wire used for data transfer
- One or more additional wires used for control (but, some protocols may not use additional control wires)
- Higher throughput for long distance communication
 - Often across processing node
- Lower cost in terms of wires (cable)
- E.g., USB, Ethernet, RS232, I²C, etc.

42



Parallel Communication

- Multiple buses used for data transfer
- One or more additional wires used for control
- Higher throughput for short distance communication
 - Data misalignment problem
 - Often used within a processing node
- Higher cost in terms of wires (cable)
- E.g., ISA, AMBA, PCI, etc.

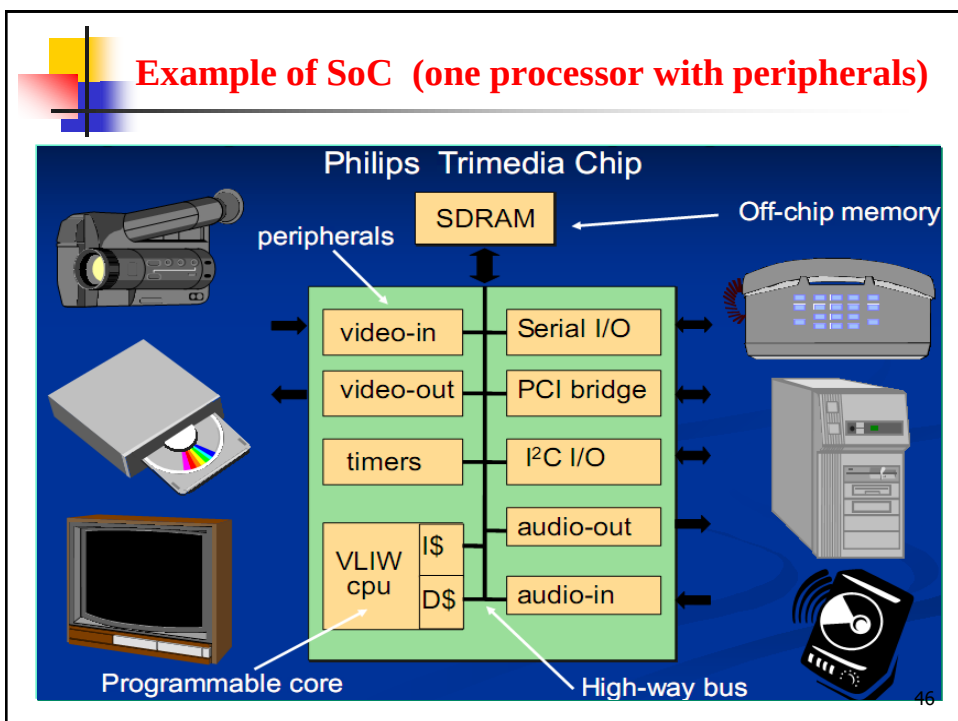
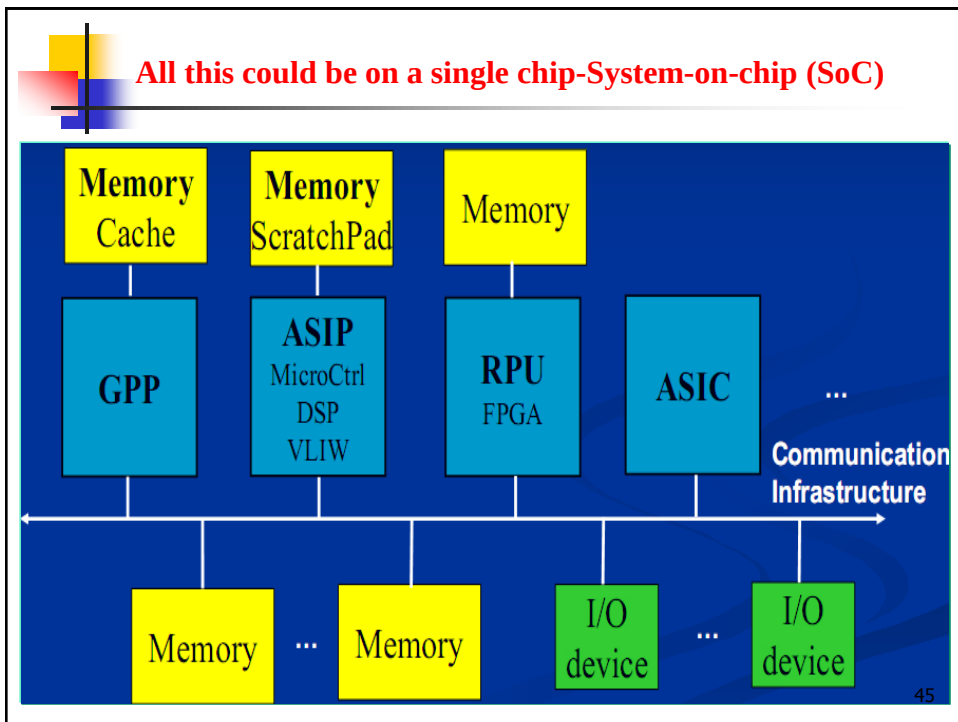
43



Wireless Communication

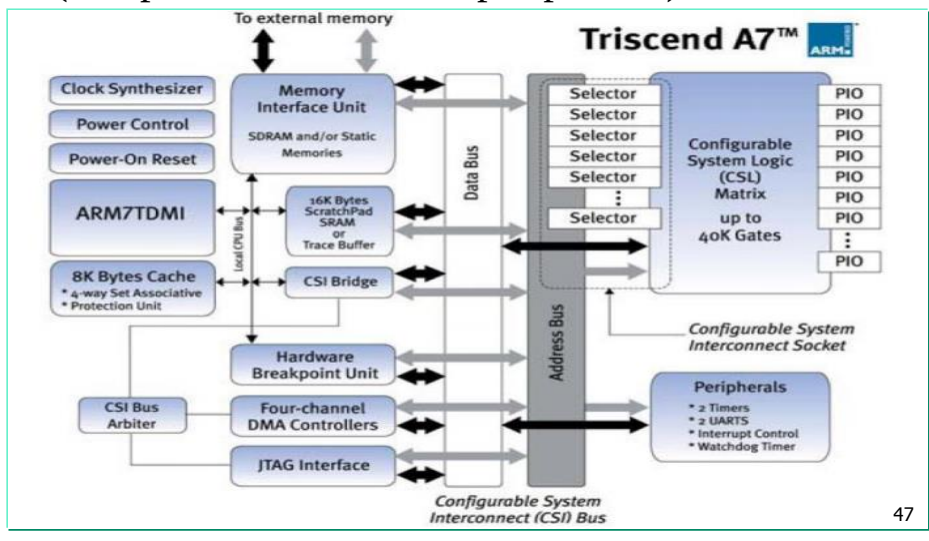
- Infrared (IR)
 - Electronic wave frequencies just below visible light spectrum
 - Diode emits infrared light to generate signal
 - Infrared transistor detects signal, conducts when exposed (tiếp xúc) to infrared light
 - Cheap to build
 - Need line of sight, limited range
- Radio frequency (RF)
 - Electromagnetic wave frequencies in radio spectrum
 - Analog circuitry and antenna needed on both sides of transmission
 - Line of sight not needed, transmitter power determines range

44



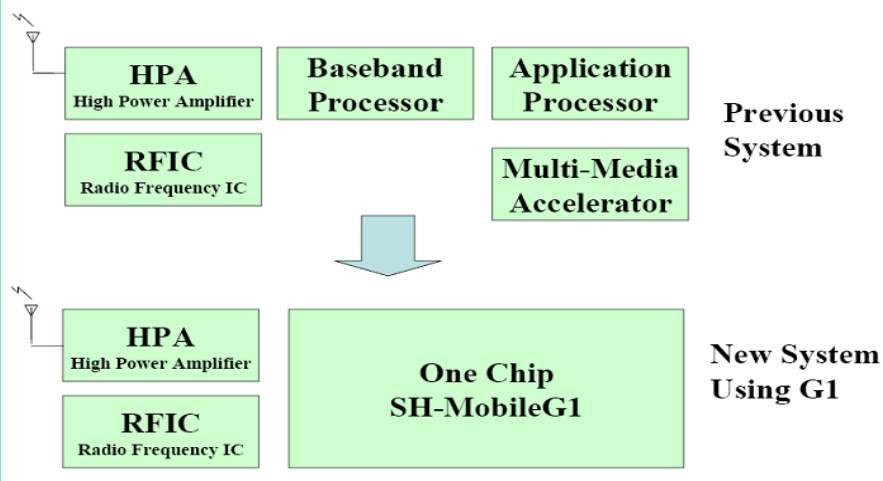
Examples of Configurable SoC

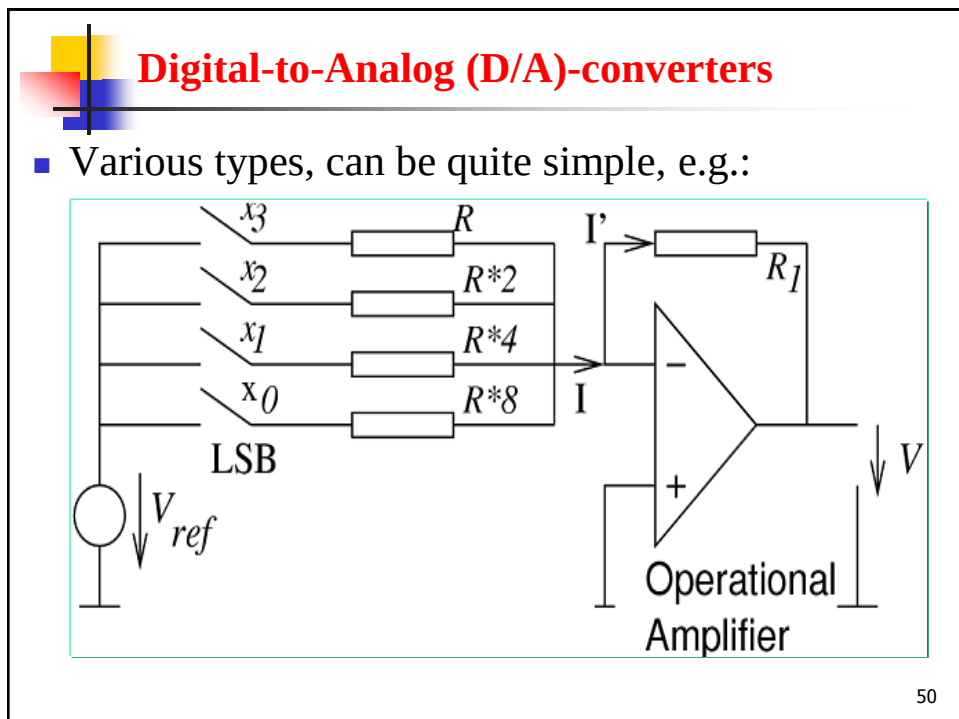
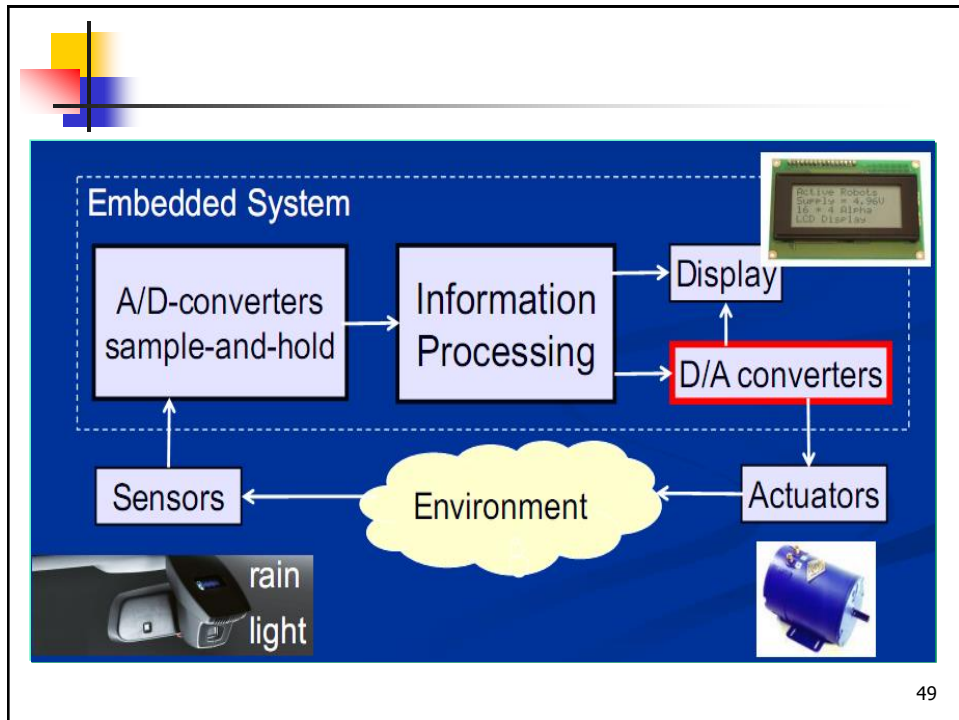
- (one processor + RPU + peripherals)



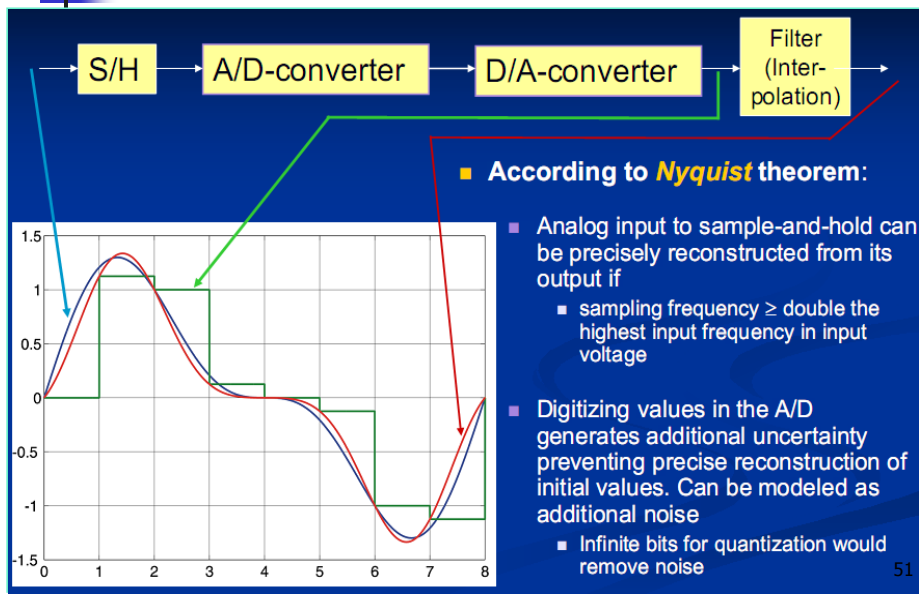
Trend: Multi-Processor SoC (MPSoC)

3G Multi-Media Cellular Phone System





Reconstruction Analog from Digital?



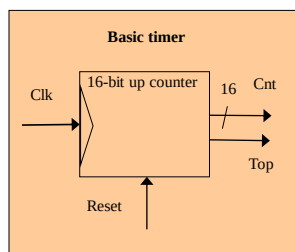
Peripherals

- Perform specific computation task
- *Custom* single-purpose processors
 - Designed by us for a unique task
- *Standard* single-purpose processors
 - “Off-the-shelf”
 - pre-designed for a common task



Timers

- Timers: measure time intervals
 - To generate timed output events
 - To measure input events
 - Top: max count reached
- Range and resolution

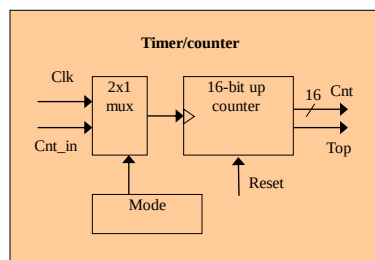


53



Counters

- Counter: like a timer, but counts pulses on a general input signal rather than clock
 - e.g., count cars passing over a sensor
 - Can often configure device as either a timer or counter



54



Watchdog Timer

- Must reset timer every X time unit, else timer generates a signal
- Common use: detect failure, self-reset

55



UART

- UART: Universal Asynchronous Receiver Transmitter
 - Takes parallel data and transmits serially
 - Receives serial data and converts to parallel
- Parity: extra bit for simple error checking
- Start bit, stop bit
- Baud rate
 - Signal changes per second
 - Bit rate, sometimes different

56



Pulse Width Modulator (PWM)

- Generates pulses with specific high/low times
- Duty cycle: % time high
 - Square wave: 50% duty cycle
- Common use: control average voltage to electric device
 - Simpler than DC-DC converter or digital-analog converter
 - DC motor speed, dimmer lights

57

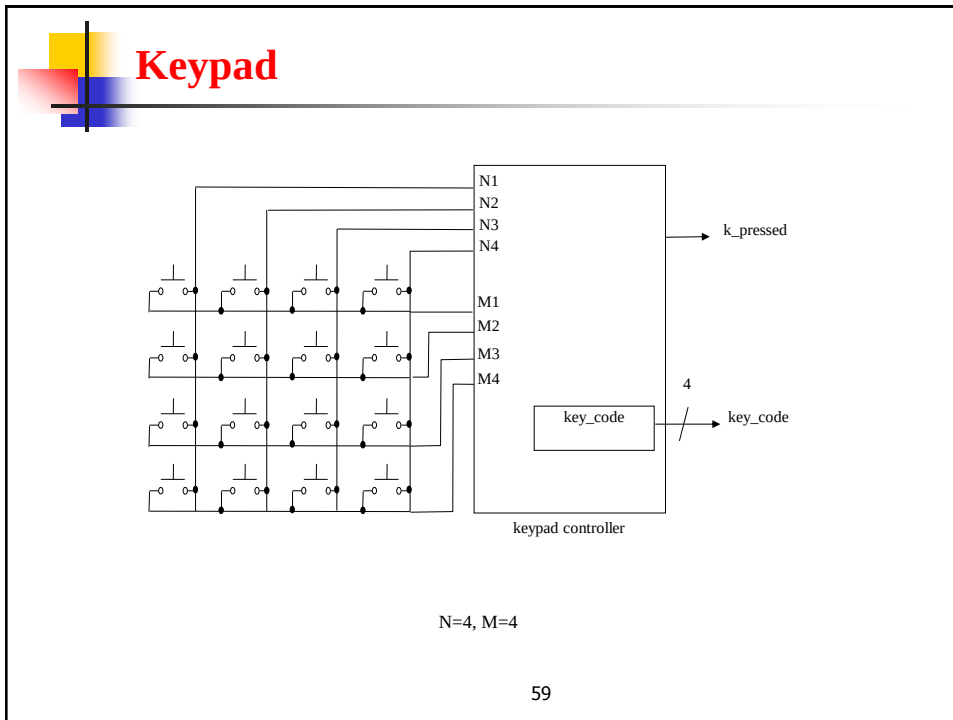


LCD

- Liquid Crystal Display
- N rows by M columns
- Controller build into the LCD module
- Simple microprocessor interface using ports
- Software controlled



58



Stepper Motor Controller

- Stepper motor: rotates fixed number of degrees when given a “step” signal
 - In contrast, DC motor just rotates when power applied, coasts to stop
- Rotation achieved by applying specific voltage sequence to coils
- Controller greatly simplifies this

60



Software

- Device driver
- RTOS (real time operating system)
- Middleware and application softwares

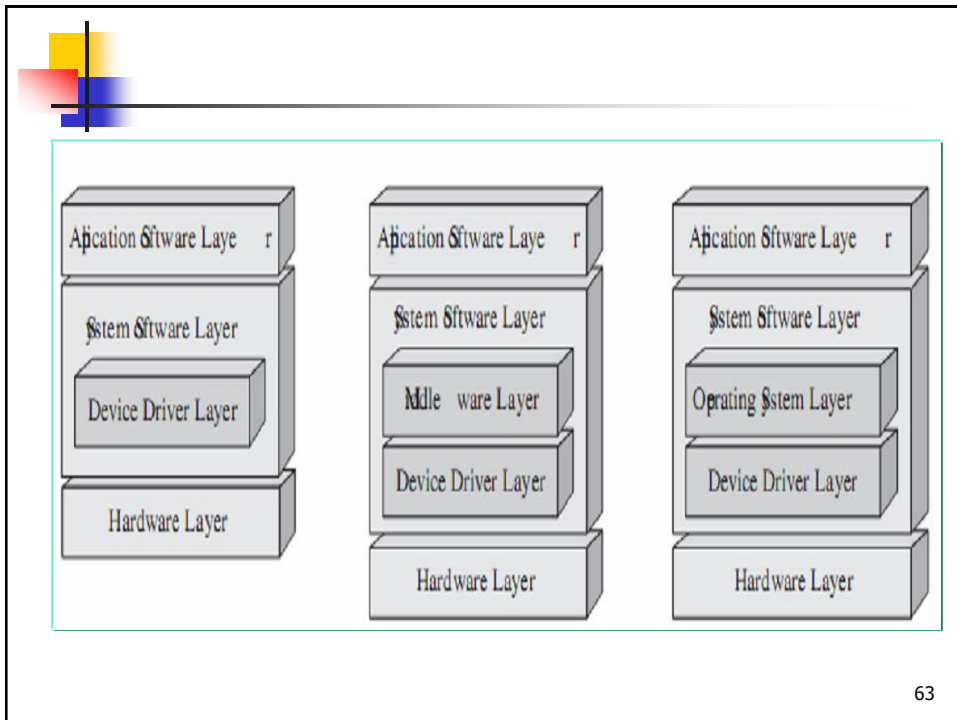
61



Device driver

- Most of embedded hardware need software to initial and mange
- The software interface and manage hardware which is called device driver
- Device driver manage accessing to hardware of layer software on higher level
- Link among hardware, OS, middleware and application layer

62



Component of driver

- Drivers are combined from the following functions
 - IO startup :

The number 64 is located in the bottom right corner of the slide frame.